Question 1 - Definitions

The transition probabilities for rolling and holding:

$P_{i,j,k,roll}$ = 1/6 (1 - $P_{j,i,0}$ + $P_{i,j,k+2}$ + $P_{i,j,k+3}$ + $P_{i,j,k+4}$ + $P_{i,j,k+5}$ + $P_{i,j,k+6}$)

$P_{i,j,k,hold}$ = 1 - $P_{j,i+k,0}$

These equations reflect the probabilistic outcomes of either choosing to roll or hold at a given game state, where i is the current player's score, j is the opponent's score, and k is the current turn total. The roll expression accounts for the possibility of losing the turn (rolling a 1), or adding the result to the turn total, while the hold expression models the probability of passing the turn and adding the accumulated turn total to the player's score.

Question 2a - Evaluating P_{99,99,0}

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

$\mathbf{P_{98,99,0} = (1/6)(1 - P_{99,98,0}) + 5/6}$

Before solving algebraically, we will simplify our expression by ruling out one of the terms in the max function.

Let $x = 1 - P_{99,98,0}$. Since $P_{99,98,0} \in [0, 1]$, we know that $x \in [0, 1]$.

Now we compare the two arguments of the max function:

First term: x
Second term: $(1/6)x + 5/6$
We determine which is larger:

$(1/6)x + 5/6 \geq x$
$\Leftrightarrow (1/6)x + 5/6 - x \geq 0$
$\Leftrightarrow (-5/6)x + 5/6 \geq 0$
$\Leftrightarrow 5/6 \geq (5/6)x$
$\Leftrightarrow 1 \geq x$

This is always true since $x \leq 1$, so:

$(1/6)(1 - P_{99,98,0}) + 5/6 \geq 1 - P_{99,98,0}$
with equality only when $P_{99,98,0} = 0$.

Thus, the maximum is always achieved by the second term, and we simplify:

$P_{98,99,0} = (1/6)(1 - P_{99,98,0}) + 5/6$

$$P_{99,99,0} = \frac{1}{6}(1 - P_{99,99,0}) + \frac{5}{6}$$

$$X = \frac{1}{6}(1-X) + \frac{5}{6}$$

$$-\frac{5}{6} \qquad\qquad -\frac{5}{6}$$

$$X - \frac{5}{6} = \frac{1}{6}(1-x)$$

$$-X \qquad\qquad -X$$

$$-\frac{5}{6} = \frac{1}{6}(1-x) - x$$

$$-\frac{5}{6} = \frac{1}{6} - \frac{1}{6}X - x$$

$$-\frac{5}{6} = \frac{1}{6} - \frac{7}{6}x$$

$$-\frac{1}{6} \qquad -\frac{1}{6}$$

$$\frac{-1}{-\frac{7}{6}} = \frac{-\frac{7}{6}x}{-\frac{7}{6}}$$

$$X = \frac{6}{7}$$

$$\boxed{P_{99,99,0} = \frac{6}{7}}$$

Question 2b - Evaluating P_{98,99,0} and P_{99,98,0}

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

$\mathbf{P_{98,99,0} = (1/6)(1 - P_{99,98,00}) + 5/6}$
$\mathbf{P_{99,98,0} = (1/6)(1 - P_{98,99,0}) + 5/6}$

Before solving algebraically, we will simplify our expressions by ruling out one of the terms in the max functions (for both expressions).

Let $x = 1 - P_{98,99,0}$. Again, since $P_{98,99,0} \in [0, 1]$, we know $x \in [0, 1]$.
(The same follows if we let $x = 1 - P_{99,98,0}$)

Compare the two arguments:

First term: x
Second term: $(1/6)x + 5/6$

We determine the larger term (Just as we did in part a):

$(1/6)x + 5/6 \geq x$
$\Leftrightarrow (1/6)x + 5/6 - x \geq 0$
$\Leftrightarrow (-5/6)x + 5/6 \geq 0$
$\Leftrightarrow 5/6 \geq (5/6)x$
$\Leftrightarrow 1 \geq x$

Which is always true. So again, the second term dominates the first (for both).

(except when $P_{98,99,0} = 0$, which cannot happen do to the constant term in our expressions: 5/6)

Hence:

$P_{98,99,0} = (1/6)(1 - P_{99,98,00}) + 5/6$
and
$P_{99,98,0} = (1/6)(1 - P_{98,99,0}) + 5/6$

$$P_{98,99,0} = \frac{1}{6}\left(1 - P_{99,98,0}\right) + \frac{5}{6}$$

$$P_{99,98,0} = \frac{1}{6}\left(1 - P_{98,99,0}\right) + \frac{5}{6}$$

$$x = \frac{1}{6}(1-y) + \frac{5}{6}$$

$$y = \frac{1}{6}(1-x) + \frac{5}{6}$$

$$x = \frac{1}{6}\left(1 - \left(\frac{1}{6}x + \frac{5}{6}\right)\right) + \frac{5}{6}$$

$$x = \frac{1}{6}\left(1 - \frac{1}{6}x - \frac{5}{6}\right) + \frac{5}{6}$$

$$x = \frac{1}{6} - \frac{1}{36}x - \frac{5}{36} + \frac{5}{6}$$

$$x = 1 - \frac{5}{36} - \frac{1}{36}x$$

$$+\frac{1}{36}x \qquad\qquad +\frac{1}{36}x$$

$$\frac{\frac{37}{36}x}{\frac{37}{36}} = \frac{\frac{31}{36}}{\frac{37}{36}}$$

$$x = \frac{1116}{1332}$$

$$x = \frac{31}{37}$$

$$y = \frac{1}{6}\left(1 - \frac{6}{37}\right) + \frac{5}{6}$$

$$y = \frac{1}{6} - \frac{6}{222} + \frac{5}{6}$$

$$y = 1 - \frac{1}{37}$$

$$y = \frac{36}{37}$$

$$\boxed{P_{98,99,0} = \frac{31}{37} \qquad P_{99,98,0} = \frac{36}{37}}$$

Question 2c - Evaluating P_{97,99,0}, P_{97,99,2}, P_{99,97,0}

Using the definition from Question 1 (and plugging in 1 for all terminal states) we get:

1. **$P_{97,99,0} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6)$**
2. **$P_{97,99,2} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + 5/6)$**
3. **$P_{99,97,0} = \max(1 - P_{97,99,0}, (1/6)(1 - P_{97,99,0}) + 5/6)$**

Before solving algebraically, we will simplify each of the expressions by ruling out one of the terms in the max function.

1. **$P_{97,99,0} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6)$**

We are comparing:

First term: $1 - P_{99,97,0}$
Second term: $(1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$
Let's assume for contradiction that the first term is the larger one.

Then:

$$1 - P_{99,97,0} > (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$$

Multiply both sides by 6:

$$6(1 - P_{99,97,0}) > (1 - P_{99,97,0}) + P_{97,99,2} + 4$$

Simplify:

$$5(1 - P_{99,97,0}) > P_{97,99,2} + 4$$

But we know that $P_{97,99,2} \leq 1$, so the right-hand side is at most 5. Then:

$$5(1 - P_{99,97,0}) > 5$$
$$\Rightarrow 1 - P_{99,97,0} > 1$$
$$\Rightarrow P_{99,97,0} < 0$$

This is a contradiction, because $P_{99,97,0} \geq 0$.

So our assumption must be false, and the second term is always greater than or equal to the first. Hence, we simplify:

$P_{97,99,0} = (1/6)(1 - P_{99,97,0}) + (1/6)(P_{97,99,2}) + 4/6$

2. $P_{97,99,2} = \max(1 - P_{99,97,0}, (1/6)(1 - P_{99,97,0}) + 5/6)$

Let $x = 1 - P_{99,97,0}$, with $x \in [0, 1]$.

Compare:

First term: x
Second term: $(1/6)x + 5/6$
As before:

$(1/6)x + 5/6 \geq x$
$\Leftrightarrow (-5/6)x + 5/6 \geq 0$
$\Leftrightarrow 1 \geq x$

So the second term is greater unless $P_{99,97,0} = 0$, in which case both are equal.

Therefore:

$P_{97,99,2} = (1/6)(1 - P_{99,97,0}) + 5/6$

3. $P_{99,97,0} = \max(1 - P_{97,99,0}, (1/6)(1 - P_{97,99,0}) + 5/6)$

Let $x = 1 - P_{97,99,0}$, with $x \in [0, 1]$.

Compare:

First term: x
Second term: $(1/6)x + 5/6$
Again:

$(1/6)x + 5/6 \geq x$
$\Leftrightarrow (-5/6)x + 5/6 \geq 0$
$\Leftrightarrow 1 \geq x$

So the maximum is achieved by the second term.

Thus:

$P_{99,97,0} = (1/6)(1 - P_{97,99,0}) + 5/6$

$$P_{97,99,0} = \frac{1}{6}\left(1 - P_{99,97,0}\right) + \frac{1}{6} P_{97,99,2} + \frac{4}{6}$$

$$P_{97,99,2} = \frac{1}{6}\left(1 - P_{99,97,0}\right) + \frac{5}{6}$$

$$P_{99,97,0} = \frac{1}{6}\left(1 - P_{97,99,0}\right) + \frac{5}{6}$$

$$\Downarrow$$

$$X = \frac{1}{6}(1-Z) + \frac{1}{6} Y + \frac{4}{6}$$

$$Y = \frac{1}{6}(1-Z) + \frac{5}{6}$$

$$Z = \frac{1}{6}(1-X) + \frac{5}{6}$$

$$X = \frac{1}{6}(1-z) + \frac{1}{6}Y + \frac{4}{6}$$

$$x = \frac{1}{6}\left(1 - \overbrace{\left(\frac{1}{6}(1-x) + \frac{5}{6}\right)}^{z}\right) + \frac{1}{6}\overbrace{\left(\frac{1}{6}\left(1 - \overbrace{\left(\frac{1}{6}(1-x) + \frac{5}{6}\right)}^{z}\right) + \frac{5}{6}\right)}^{y} + \frac{4}{6}$$

$$X = \frac{1}{6}\left(1 - \frac{1}{6} + \frac{1}{6}x - \frac{5}{6}\right) + \frac{1}{6}\left(\frac{1}{6}\left(1 - \frac{1}{6} + \frac{1}{6}x - \frac{5}{6}\right) + \frac{5}{6}\right) + \frac{4}{6}$$

$$X = \frac{1}{6}\left(\frac{1}{6}x\right) + \frac{1}{6}\left(\frac{1}{6}\left(\frac{1}{6}x\right) + \frac{5}{6}\right) + \frac{4}{6}$$

$$X = \frac{1}{6}\left(\frac{1}{6}x\right) + \frac{1}{6}\left(\frac{1}{36}X + \frac{5}{6}\right) + \frac{4}{6}$$

$$X = \frac{1}{36}X + \frac{1}{216}X + \frac{5}{36} + \frac{4}{6}$$

$$X = \frac{7}{216}X + \frac{29}{36}$$

$$+\frac{7}{216}X \qquad +\frac{7}{216}X$$

$$\frac{\frac{223}{216}X}{\frac{223}{216}} = \frac{\frac{29}{36}}{\frac{223}{216}}$$

$$X = \frac{6264}{8028}$$

$$X = \frac{174}{223}$$

$$Z = \frac{1}{6}(1-x) + \frac{5}{6}$$

$$Z = \frac{1}{6}\left(1 - \frac{174}{223}\right) + \frac{5}{6}$$

$$Z = \frac{1}{6}\left(\frac{49}{223}\right) + \frac{5}{6}$$

$$Z = \frac{49}{1338} + \frac{5}{6}$$

$$Z = \frac{49}{1338} + \frac{1115}{1338}$$

$$Z = \frac{1164}{1338}$$

$$Z = \frac{194}{223}$$

$$y = \frac{1}{6}(1-z) + \frac{5}{6}$$

$$y = \frac{1}{6}\left(1 - \frac{194}{223}\right) + \frac{5}{6}$$

$$y = \frac{1}{6}\left(\frac{29}{223}\right) + \frac{5}{6}$$

$$y = \frac{29}{1338} + \frac{5}{6}$$

$$y = \frac{29}{1338} + \frac{1115}{1338}$$

$$y = \frac{1144}{1338}$$

$$y = \frac{572}{669}$$

$$\boxed{\begin{aligned} P_{97,99,0} &= \frac{174}{223} \\ P_{97,99,2} &= \frac{572}{669} \\ P_{99,97,0} &= \frac{194}{223} \end{aligned}}$$